João Coelho

Computer Science and Engineering Student

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2nd-year Computer Science student (LEIC-A) at Instituto Superior Técnico, comfortable across C, Python and RISC-V. Looking for an internship where I can contribute from day one.

EDUCATION

LEIC-A — Computer Science and Engineering

2024 — 2028 (expected)

Instituto Superior Técnico — *Lisbon, Portugal*

- Coursework: systems programming, computer architecture, algorithms and mathematics.

PROJECTS

Invoicing & Inventory CLI

Systems Programming

Instituto Superior Técnico — C · Merge Sort · CLI

- Built an invoicing and stock-management system in C: catalogue, cart and invoice history.
- Implemented a hand-written merge sort, EAN checksum validation and wildcard product search.

Scrabble Game & Agent

Functional Programming

Instituto Superior Técnico — Python · AI Agent

- Built a full Scrabble engine in Python for human and bot players across three difficulty levels.
- Implemented board/move validation, letter-bag shuffling and a move-search agent.

16-bit Single-Cycle Processor

Computer Architecture

Instituto Superior Técnico — Logisim · ISA Design

- Designed a 16-bit single-cycle processor in Logisim on a 3-bit opcode scheme.
- Built a small custom instruction set (li, add, dot, dota).

SKILLS

Programming Python, C, Assembly (RISC-V), Prolog

Tools Git, Linux, Docker

LANGUAGES

Portuguese — Native · English — Fluent (C1) · French — Elementary (A2)